Optimized Tensor Packing and Quantitative Residuals for Selected Multiple Ergodic Averages

Dr. Muralidhara Subbarao
Technical Report Assisted by AI Models
www.integralresearch.net
Stony Brook University
Technical Report TR-Ram-OpenAI-722-M292

October 8, 2026

Abstract

We examine the October 4 manuscript *Pointwise Multiple Ergodic Averages for Mixing Transformations*, keeping its all-order mixing, triple-pointwise, and relative-structure inputs separate from local deductions. Optimizing the variance cutoff in its tensor-packing argument changes the rank-d counting exponent from $O_\epsilon(\sqrt{d}\log(2+d))$ to $O_\epsilon(\sqrt{d\log(2+d)})$. A corresponding dyadic rank schedule gives an explicit $O(\log A/\sqrt{A})$ selected-density residual, in place of the source's $O(A^{-1/4})$ residual. Neither estimate is a rate of mixing or of pointwise convergence. We also quantify the final conditional square pairing using retained physical Gram mass, with explicit martingale and pairing errors. Proper product marginals, finite tests, and finite-rank inversion are not substitutes for the actual selected-law construction. Classical Gaussian comparison, Hilbert-space projection, and covariance inequalities are credited as such. Contribution assessment: C3/5, a notable source-specific packing and selection-residual improvement with a conditional square-stability certificate, but no independent certification of the global pointwise theorem.

Contents

1 Statement, scope, and proof dependencies 2

2 Actual selected laws and the marginal obstruction 2

3 A free-cutoff Gaussian estimate 3

4 Optimized tensor packing 4

5 Selected-density residuals with faster summable ranks 6

6 A quantitative certificate for the final square 7

7 Original Ram comparison and worked checks 9

8 Current literature, advantages, and limitations 10

9 Verification and contribution boundary 10

1 Statement, scope, and proof dependencies

For an invertible probability-preserving transformation T with measurable inverse, the source claims that ordinary mixing implies, for every fixed $n \geq 2$ and fixed bounded complex tuple $(f_1, \ldots, f_n)$,

$$A_N(x) = \frac{1}{N} \sum_{k=1}^{N} \prod_{j=1}^{n} f_j(T^{jk}x) \longrightarrow \prod_{j=1}^{n} \int f_j \, d\mu \quad \text{a.e.} \tag{1}$$

The lengths are all positive integers. The exceptional null set may depend on the tuple; there is no asserted common null set for all bounded functions, no quantitative mixing rate, and no standard-space hypothesis on the original system. Coding the fixed tuple and its integer translates gives a standard factor; this is not a simultaneous coding of all measurable functions.

This report reads the complete cached manuscript, not the full proofs of its companion inputs. Thus (1) remains an attributed source claim. The following interfaces must not be silently strengthened or replaced.

1. **All-order mixing.** Source `inputs.tex:32-49` imports the implication from ordinary mixing to mixing of every finite order, from *Rokhlin's multiple-mixing problem for one transformation*. It is used for separated collections of times, not merely one fixed pair of observables.

2. **Triple pointwise input.** Lines 51–79 import smooth dyadic triple convergence for arbitrary invertible systems and distinct nonzero integer slopes, plus approximation profiles, from a companion. Lines 196–233 combine these with norm limits and length comparison to obtain all-length triple convergence. Norm convergence alone is insufficient.

3. **Relative structure.** Lines 81–192 use maximal distal factors, relative weak mixing, relative disjointness, and all-length distal pointwise convergence. The relative squared-norm replacement is proved in the source; the structural theorems themselves are imported.

Source file	Role and indispensable boundary
`selected.tex`	Failure selectors on the roof-one suspension; actual selected laws, functoriality, inverse limits; product laws for at most three slots conditioned on the *full* distal parameter.
`channels.tex`	Three output groups; proper-group independence; closure under normalized joining flows, using packing and selected densities.
`packing.tex`	Dimension-free tensor counting with sparse exceptional graphs. The adjustable-cutoff refinement is proved below.
`correlations.tex`	Minimal number of active slots, maximal number of output roles, countable energy saturation, and regrouping.
`square.tex`	Separate row and column marginal identities; saturation and parameter independence; a nonzero conditional pairing and an additional output role give the contradiction.

The introduction, input file, seven scientific sections in total, figure source, main file, bibliography, and README were literally read. Exact intervals and hashes are recorded separately. A complete manuscript read is not a complete audit of its cited theorem network.

2 Actual selected laws and the marginal obstruction

Normalize real tests to modulus at most one after expanding centered and real/imaginary parts. On $C = X \times [0,1)$ the suspension is $S^t(x,u) = (T^{\lfloor u+t \rfloor}x, \{u+t\})$. A failure set selects measurable

lengths $N_A(x,u) = c2^k$, c in a fixed finite subset $F \subset [1,2]$, $k \geq A$, with the selection independent of the phase u. Outside the failure set the source uses a deterministic length. For a flow extension $Y \to C$, the actual finite selected law is

$$\int H \, d\lambda_{Y,A} = \int_Y \frac{1}{N_A(\pi y)} \int_0^{N_A(\pi y)} H(S_Y^t y, S_Y^{2t} y, \ldots, S_Y^{nt} y) \, dt \, dm_Y(y). \tag{2}$$

Birkhoff along all lengths gives its one-slot marginals in the limit. The source uses one fixed ultrafilter, compact coding, and rectangle approximation to define λ_Y functorially on all extensions. Time translation by t_0 costs at most $2|t_0|2^{-A}$ before passage to the limit. These facts concern the measures in (2); an arbitrary joining with the same marginals is not an interchangeable replacement.

Let P_i denote the distal parameter of slot i. A conditional product statement given the parameters in a small set of slots does not imply the same statement given all P_i. The source supplies this stronger conclusion using relative disjointness. Its triple input then controls up to three slots. These are explicit steps, not consequences of a covariance matrix.

Proposition 2.1 (Exact proper-marginal obstruction)**.** *Let U, V, W be independent fair signs and $X_0 = UVW$. Every pair of output signs is independent of X_0, and the three outputs have product law. Conditioning on $X_0 = 1$ leaves every proper output marginal unchanged but makes $UVW = 1$ almost surely.*

Proof. For any fixed values of two output signs and of X_0, the third sign is uniquely determined. Each such triple has probability $1/8$, so the joint law of X_0 and the chosen pair is uniform. Conditional on $X_0 = 1$, four triples each have probability $1/4$; their triple product is one. Relative to the original product law the density is $\rho = 2\mathbf{1}_{\{UVW=1\}}$. □

This is a finite logical counterexample, not a divergent dynamical average or a counterexample to the actual source construction. It explains why the channel proof retains a full three-group density after selection. For an event of conditional probability p, its weighted density $a = p\rho$ satisfies $0 \leq a \leq 1$, $\int a = p$, and $\|a\|_2 \leq \sqrt{p}$. The unchanged pair marginals are useful; independence of all three groups after conditioning would be false.

3 A free-cutoff Gaussian estimate

All Hilbert spaces and matrices in the local packing argument are real; * is transpose. Finite-dimensional reductions are specified below. Constants $C, c > 0$ are absolute and may change between displays. All logarithms without a subscript are natural. No numerical value of these universal constants is certified by the finite tests.

Lemma 3.1 (Trace-sensitive Gaussian moment bound)**.** *Let $B_1, \ldots, B_r$ be rectangular matrices and $G = \sum_j g_j B_j$ for independent standard real Gaussians. For their symmetric dilations*

$$H_j = \begin{pmatrix} 0 & B_j \\ B_j^* & 0 \end{pmatrix}, \qquad S = \sum_j H_j^2,$$

suppose $\|S\|_{\mathrm{op}} \leq \tau$, $\operatorname{tr} S \leq 2d$, where $d \geq 1$ and $0 < \tau \leq d$. Then

$$\mathbb{E}\|G\|_{\mathrm{op}} \leq C\sqrt{\tau \log(2 + 2d/\tau)}. \tag{3}$$

The bound is independent of both ambient matrix dimensions.

Proof. Put $H = \sum_j g_j H_j$. Gaussian integration by parts and differentiation of a matrix power give, for integer $q \ge 2$,

$$\mathbb{E}\,\mathrm{tr}\,H^{2q} = \sum_j \sum_{a=0}^{2q-2} \mathbb{E}\,\mathrm{tr}(H_j H^a H_j H^{2q-2-a}).$$

Diagonalize H. Each summand is bounded above by the corresponding sum of $(H_j)_{uv}^2 |\lambda_u|^a |\lambda_v|^{2q-2-a}$. Weighted arithmetic–geometric mean and symmetry in u, v bound this by $\mathrm{tr}(H_j^2 |H|^{2q-2})$. Consequently

$$\mathbb{E}\,\mathrm{tr}\,H^{2q} \le (2q-1)\tau \mathbb{E}\,\mathrm{tr}\,H^{2q-2}, \qquad \mathbb{E}\,\mathrm{tr}\,H^2 = \mathrm{tr}\,S \le 2d,$$

and hence

$$\mathbb{E}\,\mathrm{tr}\,H^{2q} \le (2q-1)!!\,\tau^{q-1} 2d. \tag{4}$$

Since $\|H\|_{\mathrm{op}} = \|G\|_{\mathrm{op}}$, Jensen and $(2q-1)!! \le (2q)^q$ imply

$$\mathbb{E}\|G\|_{\mathrm{op}} \le \sqrt{2q\tau}\,(2d/\tau)^{1/(2q)}.$$

Choose $q = \max(2, \lceil \log(2 + 2d/\tau) \rceil)$. The last factor is at most $e^{1/2}$ and $q \le C \log(2 + 2d/\tau)$. This proves the claim, including $d = 1$. If $S = 0$, all $B_j = 0$ and the assertion is immediate. □

This is a classical Gaussian matrix-moment method, with the source's cutoff left free. Matrix Khintchine estimates and their dimension dependence are discussed by Tropp [2]; no new Gaussian comparison theorem is claimed.

Lemma 3.2 (Separated Gaussian tests)**.** *If $v_1, \dots, v_l \in \mathbb{R}^r$ have pairwise distance at least $h > 0$, then $\mathbb{E}\max_j \langle g, v_j \rangle \ge ch\sqrt{\log l}$ for $l \ge 2$.*

Proof. The classical Gaussian comparison principle compares centered processes by their increment variances. For completeness, interpolate independent copies as $Z_t = \sqrt{t}Z + \sqrt{1-t}Z'$ and differentiate the expectation of $\beta^{-1} \log \sum_j e^{\beta Z_{t,j}}$. Gaussian integration by parts gives

$$\frac{\beta}{4} \mathbb{E} \sum_{i,j} p_i p_j (\mathbb{E}(Z_i - Z_j)^2 - \mathbb{E}(Z_i' - Z_j')^2), \quad p_j \ge 0, \quad \sum p_j = 1.$$

It is nonnegative when the increment variances of Z dominate those of Z'. Let $\beta \to \infty$ and take independent Z_j' of variance $h^2/2$. The increment variance is h^2. Finally the independent-normal maximum has expectation at least $c\sqrt{\log l}$: integrating the normal density on $[t, t + 1/t]$, $t \ge 1$, gives a lower tail bound $ct^{-1}e^{-t^2/2}$; choose $t = \sqrt{\log l}$ for large l. The possible negative part of the maximum has bounded expectation, and the finitely many small $l \ge 2$ are covered by a smaller positive constant. □

4 Optimized tensor packing

Let $x_j \in H_1$, $y_j \in H_2$, $1 \le j \le m$, have norms at most one. For each family assume a simple exceptional graph of maximum degree at most L, outside which the absolute inner products are at most η. The graphs may differ. Tensor products are Hilbert tensor products.

Theorem 4.1 (Adjustable and balanced packing)**.** *For $0 < \epsilon \le 1$, $\eta \le \epsilon^2/10^4$, any orthogonal projection P of rank at most integer $d \ge 1$, and $0 < \tau \le d$, the number M of indices satisfying $\|P(x_j \otimes y_j)\| \ge \epsilon$ obeys*

$$M \le C(1+L)\epsilon^{-2}\left\{e^{(d/\tau)\log(1+80/\epsilon)} + e^{C\epsilon^{-2}\tau\log(2+2d/\tau)}\right\}, \tag{5}$$

$$M \le C(1+L)\epsilon^{-2}e^{C\epsilon^{-2}\sqrt{d\log(2+d)}}. \tag{6}$$

For rank zero, $M = 0$. All constants are independent of m, the ambient Hilbert spaces, and the particular vectors.

Proof. First, a subset A of unit vectors with the stated exceptional graph has

$$\Big\|\frac{1}{|A|}\sum_{j\in A} x_j\Big\|^2 \le \frac{1+L}{|A|} + \eta. \tag{7}$$

This follows by counting the diagonal and at most $|A|L$ ordered exceptional pairs; bound all remaining inner products above by η.

Restrict to the finite spans of the x_j and y_j. If the original range of P is not inside their tensor product, project that range into it and then orthogonally project onto the resulting span. For vectors in the tensor product this can only increase the projected norm, and its rank does not increase. We may therefore represent an orthonormal basis of the range by Hilbert–Schmidt orthonormal matrices $A_1, \dots, A_r$, $r \le d$, so that $\|P(x\otimes y)\|^2 = \sum_a |\langle x, A_a y\rangle|^2$. Set $R_1 = \sum_a A_a A_a^*$, $R_2 = \sum_a A_a^* A_a$; both have trace r.

Let E_i be the spectral projection of R_i on eigenvalues greater than τ, and $Q_i = I - E_i$. Then $\operatorname{rank} E_i \le d/\tau$. Discard indices with $\|E_1 x_j\| > \epsilon/10$ or $\|E_2 y_j\| > \epsilon/10$. The high-dimensional unit ball has an $\epsilon/40$ net of cardinality at most $(1+80/\epsilon)^{d/\tau}$, by the disjoint-ball volume argument. Inside a net cell containing discarded vectors, the norm of their average high projection is at least $\epsilon/20$ (each differs from a representative by at most $\epsilon/20$). Equation (7) bounds the cell size by $C(1+L)\epsilon^{-2}$. This proves the first term in (5), with a factor two absorbed in C.

For remaining indices put $B_a = Q_1 A_a Q_2$ and $v_j = (\langle x_j, B_a y_j\rangle)_{a=1}^r$. The tensor changes by norm at most $\epsilon/5$ under $Q_1 \otimes Q_2$; therefore an index originally counted has $\|v_j\| \ge \epsilon/2$. The map $x\otimes y \mapsto (\langle x, B_a y\rangle)_a$ is a contraction. Take a maximal $\epsilon/8$-separated set among the v_j, of cardinality l. Each cluster has mean coefficient vector of norm at least $3\epsilon/8$. Outside the union of the two exceptional graphs, $|\langle x_j \otimes y_j, x_{j'} \otimes y_{j'}\rangle| \le \eta^2$; that union has degree at most $2L$. Applying (7) to the uncompressed tensors and then the contraction bounds each cluster size by $C(1+L)\epsilon^{-2}$.

For $G = \sum_a g_a B_a$, the associated dilation variance satisfies $\|S\|_{\mathrm{op}} \le \tau$, $\operatorname{tr} S \le 2d$: for instance $\sum_a B_a B_a^* \preceq Q_1 R_1 Q_1 \preceq \tau Q_1$. For the separated representatives, Lemmas 3.1 and 3.2 give

$$c\epsilon\sqrt{\log l} \le \mathbb{E}\max_j \langle x_j, G y_j\rangle \le \mathbb{E}\|G\|_{\mathrm{op}} \le C\sqrt{\tau\log(2+2d/\tau)}.$$

The case $l = 1$ needs no estimate. This proves the second term.

Finally put $\ell = \log(2+d)$ and $\tau = \sqrt{d/\ell}$. For $d \ge 1$, $0 < \tau \le d$, $d/\tau = \sqrt{d\ell}$, and $\log(2+d/\tau) \le 2\ell$. Indeed $\ell \le d+2$ and $\sqrt{d\ell} \le d+2$, so $2+\sqrt{d\ell} \le (d+2)^2$. Also $\log(2+2d/\tau) \le \log 2 + 2\ell \le 3\ell$. Since $\log(1+80/\epsilon) \le C\epsilon^{-2}$ on $(0,1]$, both exponents in (5) are bounded by $C\epsilon^{-2}\sqrt{d\ell}$, proving (6). □

Corollary 4.2 (Admissible growing ranks)**.** *Fix $\epsilon > 0$, then $\eta \le \epsilon^2/10^4$, and then a finite L. If integer ranks $d_m \ge 1$ satisfy*

$$d_m \log(2+d_m) = o((\log m)^2), \tag{8}$$

then $m^{-1}\sum_{j\le m}\|P_m(x_j\otimes y_j)\| \le \epsilon + o(1)$ for every sequence of rank-d_m projections. A fixed omitted initial set has no effect on this conclusion.

Proof. Use the threshold ϵ and the bound one on all projected norms. The exceptional count divided by m tends to zero by (6). A fixed initial set contributes at most its cardinality divided by m. □

The source chooses $\tau = \sqrt{d}$, giving the larger exponent $C\epsilon^{-2}\sqrt{d}\log(2+d)$. Our gain is a cutoff optimization of that proof, not an assertion of a historically optimal packing theorem.

5 Selected-density residuals with faster summable ranks

We now match the local hypotheses to `channels.tex:158-274`. Fix the channel, its rational speeds, and an integer h clearing their denominators. At fixed phase and fractional time, the three groups give unit vectors $u_l \in L^2(\Omega_1)$, $v_l \in L^2(\Omega_2)$, $w_l \in L^2(\Omega_3)$, on their unselected product probability space. Each group has a centered occurrence. The source's imported all-order mixing, applied to finitely many offsets, gives for every fixed $\eta > 0$ a fixed initial omission and exceptional graphs of finite degree $L(\eta)$ for u_l and v_l. A bad pair solves one of finitely many bounded-distance relations between distinct positive linear time forms. There is no quantitative bound here on $L(\eta)$ or on the initial omission.

Lemma 5.1 (Projection and selected-density costs)**.** *Let $z_l = u_l \otimes v_l$, $1 \le l \le m$, and let P retain the largest $d \ge 1$ eigenvalues of $\sum_{l\le m} z_l z_l^*$, or the whole range if it has smaller dimension. Write $z_l' = (I-P)z_l$ and $T' = m^{-1}\sum_l z_l' \otimes w_l$. Then $\|T'\|_2 \le d^{-1/2}$. If an event has probability p, weighted density $a = p\rho \le 1$, and unchanged (Ω_1,Ω_2) and Ω_3 marginals, its projected and residual contributions to the trilinear average are bounded by*

$$p\frac{1}{m}\sum_l \|Pz_l\|_2 \quad \text{and} \quad \sqrt{p/d}, \tag{9}$$

respectively. Independence of the three groups under ρ is unnecessary.

Proof. The unnormalized covariance has trace at most m; its residual operator norm is at most m/d. The residual Gram matrix $A = (\langle z_l', z_{l'}'\rangle)$ has the same nonzero eigenvalues. For $B = (\langle w_l, w_{l'}\rangle)$, positivity and $\operatorname{tr} B \le m$ give

$$\|T'\|_2^2 = m^{-2}\operatorname{tr}(AB) \le m^{-2}\|A\|_{\mathrm{op}}\operatorname{tr} B \le d^{-1}.$$

For the projected term apply Cauchy–Schwarz under ρ separately to $(Pz_l)(\omega_1,\omega_2)$ and $w_l(\omega_3)$. The unchanged marginals give their original L^2 norms. For the residual use $|\int aT'| \le \|a\|_2\|T'\|_2 \le \sqrt{p/d}$. Zero-probability events are omitted. These are integrals under precisely the specified measures. □

Theorem 5.2 (An explicit dyadic residual certificate)**.** *At selector level $k \ge 1$ use, for each $c \in F$,*

$$d_k = \left\lceil \frac{k^2}{(1+\lfloor \log_2 k\rfloor)^2} \right\rceil, \qquad m_{k,c} = \lfloor c2^k/h\rfloor. \tag{10}$$

Under the channel assumptions just stated, its total projected contribution over selected levels $k \ge A$ is at most $\epsilon + o_A(1)$, after fixing ϵ, then $\eta \le \epsilon^2/10^4$, then the resulting initial omission and L. The remaining density contribution is at most

$$\Big[|F|\,2^{-K}(2K^2+8K+12)\Big]^{1/2}, \qquad K = \lfloor \log_2 A\rfloor, \quad A \ge 2. \tag{11}$$

It is $O_F(\log A/\sqrt{A})$. The source's time truncation error $O(h2^{-A})$ is additional and unchanged.

Proof. For fixed h, c, $\log m_{k,c} \asymp k$, whereas $d_k \asymp k^2/(\log k)^2$. Thus

$$\frac{d_k \log(2+d_k)}{(\log m_{k,c})^2} = O(1/\log k) \longrightarrow 0.$$

Corollary 4.2 applies uniformly over the finite set of c's and source offsets. If $p_{k,c}$ are the conditional event probabilities, $\sum_{k,c} p_{k,c} = 1$. Hence the sum of the projected bounds in (9) is at most $\epsilon + \sup_{k \geq A, c} o_k(1)$. For the residual, Cauchy–Schwarz gives

$$\sum_{k \geq A, c} \sqrt{p_{k,c}/d_k} \leq \left(|F| \sum_{k \geq A} d_k^{-1} \right)^{1/2}.$$

For $2^r \leq k < 2^{r+1}$, $1/d_k \leq (r+1)^2/2^{2r}$. Enlarging the first partial block to a full block, the sum is bounded by

$$\sum_{r \geq K} \frac{(r+1)^2}{2^r} = 2^{-K}(2K^2 + 8K + 12). \tag{12}$$

The identity follows by differentiating the geometric series twice, or by its first-order tail recurrence. Since $2^K \leq A < 2^{K+1}$, the stated asymptotic estimate follows. □

A continuous family of allowable choices. For any fixed $b > 0$, choose, at sufficiently large m,

$$d_m \asymp (\log m)^2/(\log\log m)^{1+b}.$$

These ranks satisfy (8); defining the finitely many small ranks as one removes any logarithmic singularity. Their dyadic reciprocal tail is $O_b((\log A)^{1+b}/A)$, by a dyadic-block comparison or integration, so the residual is $O_b(A^{-1/2}(\log A)^{(1+b)/2})$. The exact schedule (10) realizes $b = 1$ without real logarithms in the rank calculation. The old exponent at $d_k \asymp k^2/(\log k)^2$ is of order k, not $o(k)$; that estimate alone therefore cannot ensure an exceptional fraction tending to zero for every fixed ϵ. The new exponent is $O(k/\sqrt{\log k})$ and can.

Limits of the conclusion. After taking $A \to \infty$ with the indicated data fixed, one lets $\epsilon \downarrow 0$. Choosing $\epsilon = \epsilon_A \downarrow 0$ inside the estimate is unjustified without uniform control of the mixing thresholds. Thus the explicit residual in (11) is not an effective channel-closure modulus, a mixing rate, or a rate for (1). The initial omission, graph degree, and finite-channel data may all deteriorate. This is the same logical order used by the source.

6 A quantitative certificate for the final square

The source does not exchange rows and columns by symmetry. In its joining extension Z, the old row has law λ_Y by a one-slot marginal; the column has that law by functoriality. Saturation gives the conditional column identity after adjoining the entire parameter and a row. A separate deterministic-parameter argument makes the column parameter p and the row parameter e independent given a common factor w. These distinctions in `square.tex:64-164` are retained here.

The following theorem isolates a quantitative consequence of those identities. It also states exactly what approximate conditional expectations would have to control.

Theorem 6.1 (Retained-mass square stability)**.** *Let w have probability law ζ. On a measurable field of separable real Hilbert spaces H_w, let $p \sim \alpha_w$ and $e \sim \beta_w$ be independent conditional on w. Suppose a measurable map R pushes β_w to α_w. Let $G_w(p), U_w(e) \in H_w$ have norms at most one. Assume*

$$\int \mathbb{E}_{p\sim\alpha_w} \|\mathbb{E}[U_w(e) \mid R(e) = p] - G_w(p)\|^2 \, d\zeta(w) \le \xi^2. \tag{13}$$

For measurable orthogonal projections P_w of rank at most $r \ge 1$, put

$$M = \int \mathbb{E}\|P_w G_w(p)\|^2 \, d\zeta(w), \qquad Z(w,p,e) = \langle G_w(p), U_w(e)\rangle.$$

Then

$$\mathbb{E}|Z| \;\ge\; \mathbb{E}|Z|^2 \;\ge\; (M/\sqrt{r} - \xi)_+^2. \tag{14}$$

If a row-column conditional pairing V obeys $\mathbb{E}|V - Z| \le \rho$, there is a measurable parameter sign Θ, $|\Theta| \le 1$, with $\mathbb{E}(\Theta V) \ge (M/\sqrt{r} - \xi)_+^2 - \rho$. In the exact tower case, if $0 \le t^2 < M^2/r$ and $t < 1$, then

$$\Pr\{|Z| > t\} \ge \frac{M^2/r - t^2}{1 - t^2}. \tag{15}$$

Proof. First set $\xi = 0$. Define the positive trace-class covariance operator $\Sigma_w = \mathbb{E}_p[G_w(p) \otimes G_w(p)]$. This is an uncentered second-moment operator; subtracting a mean is not part of the formula. Jensen conditional on $p, R(e)$ and the product parameter law give

$$\mathbb{E}_{p,e}|\langle G_w(p), U_w(e)\rangle|^2 \ge \mathbb{E}_{p,p'}|\langle G_w(p), G_w(p')\rangle|^2 = \|\Sigma_w\|_{\mathrm{HS}}^2.$$

The equality follows first on finite orthonormal expansions, then by Tonelli and bounded L^2 approximation. Compression and the eigenvalue Cauchy–Schwarz inequality yield

$$\|\Sigma_w\|_{\mathrm{HS}}^2 \ge \|P_w \Sigma_w P_w\|_{\mathrm{HS}}^2 \ge \frac{(\mathrm{tr}(P_w \Sigma_w P_w))^2}{r}.$$

Integrating and Jensen in w gives $\mathbb{E}|Z|^2 \ge M^2/r$.

For general ξ, let the tower error in (13) be $b_w(p')$. Conditional Jensen bounds $\|Z\|_{L^2}$ below by the L^2 norm of $\langle G_w(p), G_w(p') + b_w(p')\rangle$ under the conditional product law. The error term has L^2 norm at most ξ, since $\|G\| \le 1$. The triangle inequality therefore gives $\|Z\|_2 \ge (M/\sqrt{r} - \xi)_+$. Also $|Z| \le 1$, so $\mathbb{E}|Z| \ge \mathbb{E}|Z|^2$. Choosing $\Theta = \mathrm{sign}(V)$ and using $\mathbb{E}|V| \ge \mathbb{E}|Z| - \rho$ proves the pairing statement. Finally, $\mathbb{E}|Z|^2 \le t^2 + (1 - t^2)\Pr(|Z| > t)$ proves (15). □

Corollary 6.2 (Correlation and a certified tail)**.** *In the source square let $q = \prod_{j\ne b} h_j(L_j(y_j))$, $g(P, y_b) = \mathbb{E}[q \mid P, y_b]$, and $u(e, y_b) = \mathbb{E}[q \mid e, y_b]$. Use the source's conditional fibre $H_w = L^2(m_{Y,w}; \mathbb{R})$ and parameter laws, so the exact tower in Theorem 6.1 holds. If the original unit-bounded correlation has magnitude $c > 0$ and a measurable rank-r family satisfies $\int \mathbb{E}\|(I - P_w)G_w\|^2 \, d\zeta \le \delta$, then*

$$\mathbb{E}|Z| \ge (c^2 - \delta)_+^2/r.$$

With tower error ξ and pairing error ρ, the certified signed correlation is at least $((c^2 - \delta)_+/\sqrt{r} - \xi)_+^2 - \rho$.

Proof. Conditional expectation and Cauchy–Schwarz bound the original correlation by $\|g\|_2$, so $\int \mathbb{E}\|G\|^2 \ge c^2$. Orthogonal decomposition gives $M \ge (c^2 - \delta)_+$, and the theorem applies. □

No finite rank uniformly captures every possible g. The source uses separability to obtain a qualitative contradiction; this report's finite certificate is conditional on a retained mass or an actual tail bound. Saturation is an infinite energy-maximizing extension construction, not a finite algorithm with known complexity. The source maximizes output roles only after minimizing active slots; proper-marginal identities alone do not supply either extremality or the square's conditional identities.

7 Original Ram comparison and worked checks

Fresh original proof passages in TR248 [6], lines 133–267, 349–419, and 1030–1121, supply the relevant comparison: finite physical coordinates, exact projection leakage, and a matrix perturbation bound. They do not prove selected-law identities. For a complete patch bank with $\sum_p w_p^2 = 1$, local orthogonal projections Π_p, and multiplication operators M_p, the original proof gives

$$F^*F = \sum_p M_p \Pi_p M_p = I - L, \qquad \langle Lf, f\rangle = \sum_p \|(I - \Pi_p)(w_p f)\|^2. \tag{16}$$

Indeed local Parseval expands $w_p f$, and summing $\|w_p f\|^2$ returns $\|f\|^2$. Subtracting retained terms proves (16). The omitted quantity is $w_p f$, not the unrestricted signal on a patch. This is the appropriate analogue of the retained mass M and tail δ above, not a claim that finite coefficients determine the entire fibre.

For nonorthonormal coordinates with physical Gram matrix $H \succ 0$, the coefficient norm is a^*Ha, and a raw analysis matrix B represents $BH^{-1/2}$ in orthonormal physical coordinates. On a finite probability partition the Gram is the diagonal matrix of atom probabilities, after removing zero-mass atoms. For raw local atoms $e^{st}t^j$, original TR248 derives $H_{ij} = \int e^{2\Re(s)t} t^{i+j}\, d\mu(t)$ by differentiating the exponential moment on a bounded patch. It explicitly requires positive definiteness of the full retained Gram, not just a positive zeroth moment. These are finite classical coordinate identities with stated measures. An exact inverse in such a model says nothing about its orthogonal complement.

The later private comparative report TR-item-313 [7], read in full, emphasizes the proper-marginal and moving-observable obstructions. Selected passages of TR-item-168 [8] distinguish norm limits, genuine oscillation control, and bounded Cesaro gap filling. These are comparisons, not foundational proofs of the source's two companion claims. No earlier M280, M294, or M295 report is used as a proof dependency here.

An exact square example. In $H = \mathbb{R}^2$, let G be uniform on $\{e_1, -e_1, e_2, -e_2\}$ and let U be an independent copy whose parameter maps identically to the law of G. The tower is exact, $\Sigma = I/2$, and $\mathbb{E}|\langle G, U\rangle|^2 = \mathbb{E}|\langle G, U\rangle| = 1/2$. With $P = I$ the retained-mass bound $M^2/r = 1/2$ is exact. With projection onto one axis, $M = 1/2$ and the bound becomes $1/4$, showing the cost of discarding a direction. This is a finite fixture, not a realized selected dynamical joining.

A dyadic certificate. For $A = 2^{20}$ and $|F| = 1$, (11) has squared bound $972/2^{20}$; its square root is less than $31/1000$. This is solely the density-residual bound. The uncontrolled projected term must still be handled in the correct order of limits. Ranks in (10) use exact integer arithmetic, and the geometric tail identity is exact rather than a floating-point extrapolation.

8 Current literature, advantages, and limitations

Public primary searches were made on October 8, 2026, using only public titles and URLs. Ryzhikov's 2024 survey abstract [3] states the general multiple-mixing problem as open at that writing; the July 2026 abstract [4] concerns Kalikow's rank-one theorem and related joinings, not an all-system replacement for the source's imported claim. The September 2, 2026 revision of Kosz–Mirek–Peluse–Wan–Wright [5] establishes pointwise polynomial averages for distinct degrees. The times $k, 2k, \dots, nk$ all have degree one, so that hypothesis does not cover (1). These scoped primary checks neither certify nor refute the new cached manuscript. Abstract metadata are not full proof audits; the public source's provenance is likewise not a theorem certificate.

Advantages. The packing refinement is dimension-free and retains the source's exact exception-graph assumptions. Its larger admissible ranks produce a genuinely smaller summable density residual. The square certificate converts a positive, physically normalized finite mass into an explicit nonzero correlation margin and tolerates declared tower and pairing errors. No independence assumption has been added to the selected full three-group law.

Costs and nonclaims. The packing constants are universal but not numerically optimized. Rank projections are analytical objects; no fast construction in a general infinite Hilbert space is supplied. Larger ranks can increase computational cost. An empirical Gram matrix without a tail certificate does not certify the full conditional fibre. No finite probability-space calculation proves all-order mixing, triple-pointwise convergence, saturation, or the actual selected-law closure on every extension. In particular this report does not announce a new global mixing or pointwise ergodic theorem.

Further work. One could optimize (5) for a specified ϵ and variance profile instead of using one universal cutoff. Actual quantitative multi-correlation bounds would be needed to choose ϵ as a function of the selector level. In concrete systems, conditional finite partitions with proved leakage bounds could make Corollary 6.2 effective. The principal unresolved audit tasks remain the imported companion proofs and their interfaces, not a rebranding of physical Gram matrices.

9 Verification and contribution boundary

The owned exact checker uses rational arithmetic for finite exceptional-graph identities, Gaussian trace-moment fixtures, parity marginals, the dyadic tail recurrence and residual bounds, and covariance compression examples. It records test cases separately from individual assertions: 6931 cases and 18082 assertions passed in the author run. No Monte Carlo experiment or finite search is offered as a proof of the all-dimensional packing theorem or the dynamical statement. The manuscript proofs supply the quantified analytic arguments; executable checks exercise their finite algebra and boundary cases. Parent checks, if any, are separate evidence and are not included in the author-run counts.

The literal-read record covers all 2035 lines in 11 cached public texts, including 1724 scientific-section lines. It separately identifies the full 360-line later TR-item-313 report, the selected TR-item-168 passages, and 298 freshly read original TR248 lines. Unread complements and pre/post hashes are recorded; there is no whole-corpus or full external-proof claim. Native compilation is attempted before the standard export fallback. Final compilation, exact-check totals, and individual page inspection are recorded in the owned build and evidence files.

The C3 assessment rests on Theorem 4.1 and its source-matched residual consequence, with Theorem 6.1 as a secondary explicit stability certificate. It does not count elementary parity, Parseval, Gaussian comparison, or a renamed covariance matrix as independent novelty. Applicability L3 remains a separate catalogue designation.

References

[1] OpenAI, *Pointwise Multiple Ergodic Averages for Mixing Transformations*, October 4, 2026, cached manuscript at pinned commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Public source directory.

[2] J. A. Tropp, *Second-Order Matrix Concentration Inequalities*, Appl. Comput. Harmon. Anal. 44 (2018), 700–736. Author-hosted preprint; journal DOI.

[3] V. V. Ryzhikov, *Multiple mixing, 75 years of Rokhlin's problem*, arXiv:2411.07234v2 (November 12, 2024). Primary abstract.

[4] V. V. Ryzhikov, *Around Kalikow's theorem on multiple mixing*, arXiv:2607.11852v2 (July 14, 2026). Primary abstract.

[5] D. Kosz, M. Mirek, S. Peluse, R. Wan, and J. Wright, *The multilinear circle method and a question of Bergelson*, arXiv:2411.09478v4 (September 2, 2026). Primary abstract and metadata.

[6] M. Subbarao, *Stable Adaptive Ram Frames and Task-Aware Approximation*, TR248, original private corpus, locally consulted proof passages 133–267, 349–419, 1030–1121. Not uploaded or publicly searched.

[7] M. Subbarao, TR-item-313-astra, private comparative report, 360 lines, read in full locally. Later comparison, not an original proof of the all-order mixing implication.

[8] M. Subbarao, TR-item-168-astra, private comparative report, locally read lines 1–117, 216–272, 347–470, and 530–577. Remaining passages not claimed read in this assignment.